Supply Chain Data Analysis Report

SQL Data Cleaning and View Creation

The following script initializes the environment and creates a cleaned view of the supply chain data, standardizing date formats and calculating key performance metrics such as lead time variance.

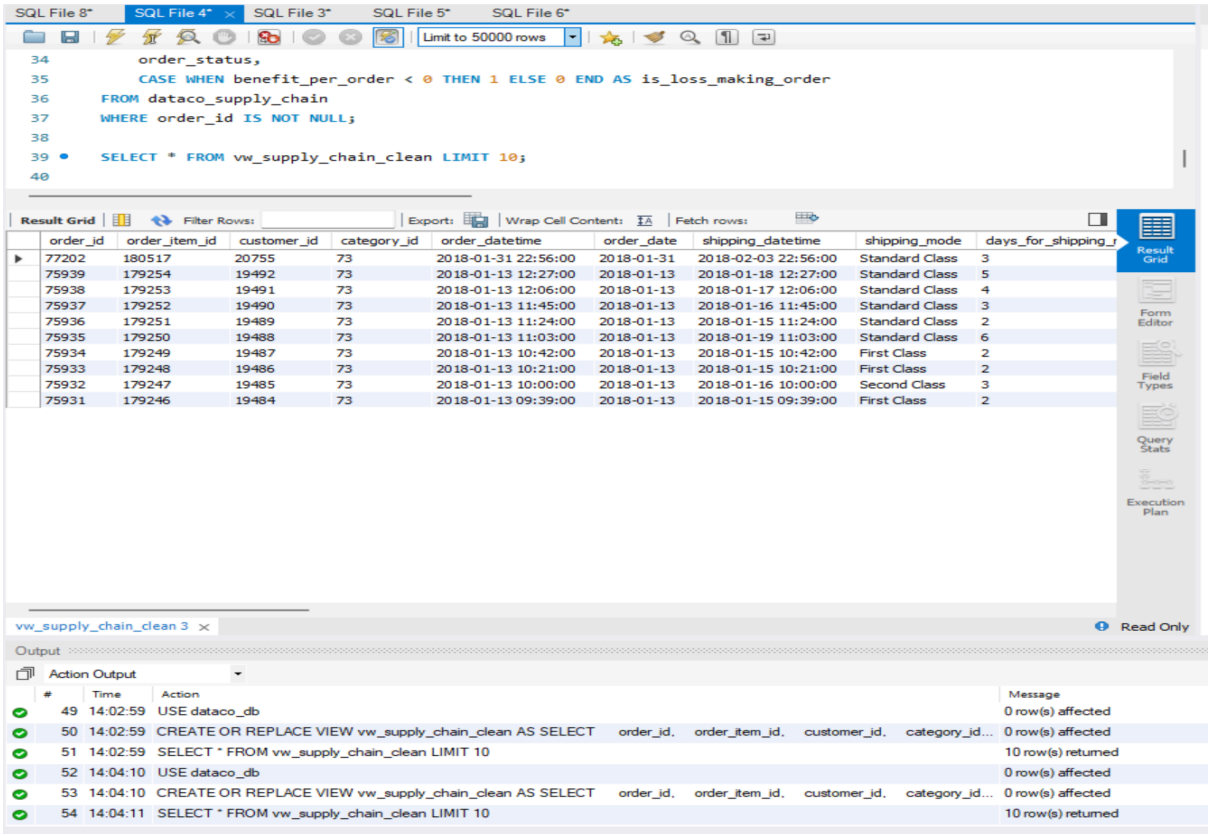
```sql
USE dataco_db;

CREATE OR REPLACE VIEW vw_supply_chain_clean AS
SELECT
    order_id,
    order_item_id,
    customer_id,
    category_id,
    STR_TO_DATE(order_date, '%c/%e/%Y %H:%i') AS order_datetime,
    CAST(STR_TO_DATE(order_date, '%c/%e/%Y %H:%i') AS DATE) AS
order_date,
    STR_TO_DATE(shipping_date, '%c/%e/%Y %H:%i') AS
shipping_datetime,
    shipping_mode,
    days_for_shipping_real,
    days_for_shipment_scheduled,
    (days_for_shipping_real - days_for_shipment_scheduled) AS
lead_time_variance_days,
    delivery_status,
    late_delivery_risk,
    market,
    order_region,
    order_country,
    order_city,
    customer_segment,
    customer_country,
    customer_city,
    category_name,
    product_name,
    product_price,
    order_item_quantity,
```

```sql
    sales AS gross_sales_usd,
    order_item_discount AS discount_amount_usd,
    order_item_discount_rate AS discount_rate,
    benefit_per_order AS net_profit_usd,
    order_item_profit_ratio AS profit_ratio,
    order_status,
    CASE WHEN benefit_per_order < 0 THEN 1 ELSE 0 END AS
is_loss_making_order
FROM dataco_supply_chain
WHERE order_id IS NOT NULL;
```

Data Preview

A selection of records from the cleaned view to verify data integrity and schema mapping.

```sql
SELECT * FROM vw_supply_chain_clean LIMIT 10;
```



Analysis Queries

Shipping Performance by Region

This query evaluates late delivery rates and average shipping delays across different regions and shipping modes.

```sql
SELECT
    order_region,
    shipping_mode,
    COUNT(order_id) AS total_orders,
    ROUND(AVG(days_for_shipping_real), 2) AS
avg_actual_shipping_days,
    ROUND(AVG(days_for_shipment_scheduled), 2) AS
avg_scheduled_shipping_days,
    ROUND(AVG(days_for_shipping_real -
days_for_shipment_scheduled), 2) AS avg_lead_time_variance_days,
    ROUND(SUM(late_delivery_risk) / COUNT(order_id) * 100, 2) AS
late_delivery_rate_pct
FROM dataco_supply_chain
GROUP BY order_region, shipping_mode
ORDER BY late_delivery_rate_pct DESC
LIMIT 15;
```

Profitability Analysis by Category

Identifies product categories with the highest volume of loss-making orders and calculates the direct financial impact.

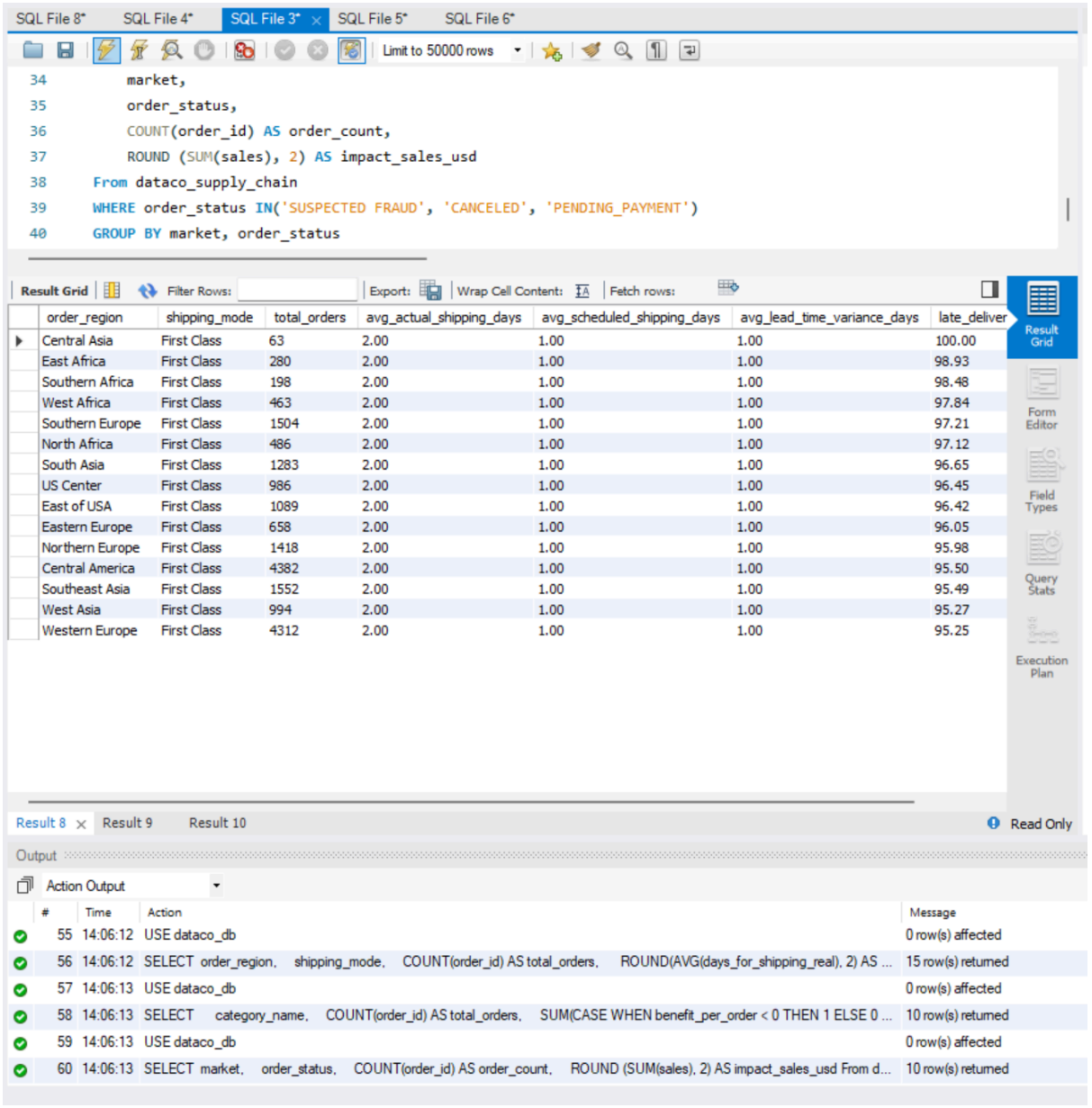
```sql
SELECT
    category_name,
    COUNT(order_id) AS total_orders,
    SUM(CASE WHEN benefit_per_order < 0 THEN 1 ELSE 0 END) AS
loss_making_orders,
    ROUND(SUM(CASE WHEN benefit_per_order < 0 THEN 1 ELSE 0 END)
/ COUNT(order_id) * 100, 2) AS loss_order_pct,
    ROUND(SUM(CASE WHEN benefit_per_order < 0 THEN
benefit_per_order ELSE 0 END), 2) AS total_direct_losses_usd,
```

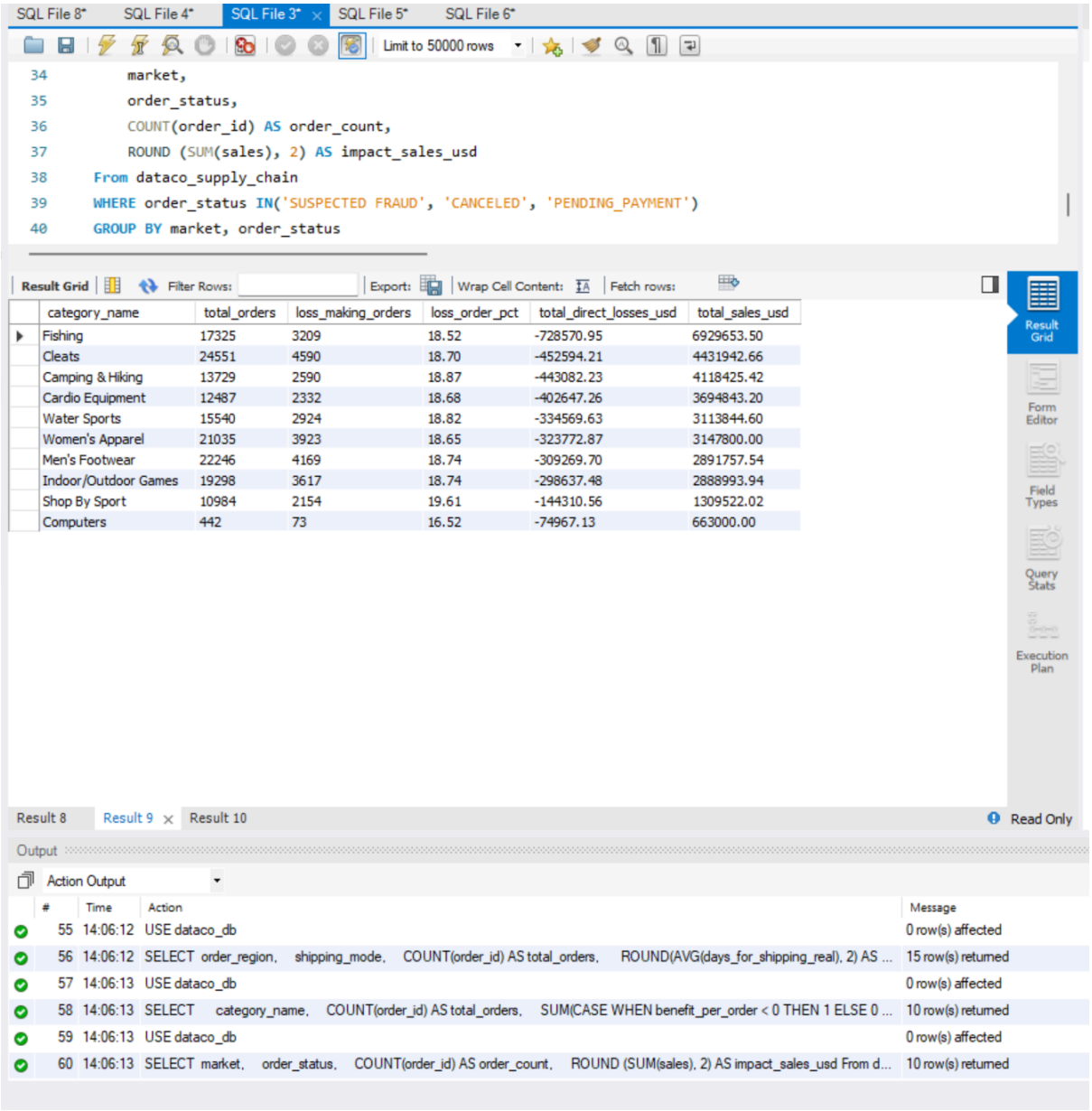
```
    ROUND(SUM(sales), 2) AS total_sales_usd
FROM dataco_supply_chain
GROUP BY category_name
HAVING total_direct_losses_usd < 0
ORDER BY total_direct_losses_usd ASC
LIMIT 10;
```

Financial Impact of Fraud and Cancellations

Analyzes the distribution of sales impact based on high-risk order statuses across different global markets.

```SQL
SELECT
    market,
    order_status,
    COUNT(order_id) AS order_count,
    ROUND (SUM(sales), 2) AS impact_sales_usd
From dataco_supply_chain
WHERE order_status IN('SUSPECTED FRAUD', 'CANCELED',
'PENDING_PAYMENT')
GROUP BY market, order_status
ORDER BY impact_sales_usd DESC;
```


SQL File 8*
SQL File 4*
SQL File 3*
SQL File 5*
SQL File 6*
Limit to 50000 rows
34 market,
35 order_status,
36 COUNT(order_id) AS order_count,
37 ROUND (SUM(sales), 2) AS impact_sales_usd
38 From dataco_supply_chain
39 WHERE order_status IN('SUSPECTED FRAUD', 'CANCELED', 'PENDING_PAYMENT')
40 GROUP BY market, order_status
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Fetch rows:
order_region | shipping_mode | total_orders | avg_actual_shipping_days | avg_scheduled_shipping_days | avg_lead_time_variance_days | late_deliver
Central Asia | First Class | 63 | 2.00 | 1.00 | 1.00 | 100.00
East Africa | First Class | 280 | 2.00 | 1.00 | 1.00 | 98.93
Southern Africa | First Class | 198 | 2.00 | 1.00 | 1.00 | 98.48
West Africa | First Class | 463 | 2.00 | 1.00 | 1.00 | 97.84
Southern Europe | First Class | 1504 | 2.00 | 1.00 | 1.00 | 97.21
North Africa | First Class | 486 | 2.00 | 1.00 | 1.00 | 97.12
South Asia | First Class | 1283 | 2.00 | 1.00 | 1.00 | 96.65
US Center | First Class | 986 | 2.00 | 1.00 | 1.00 | 96.45
East of USA | First Class | 1089 | 2.00 | 1.00 | 1.00 | 96.42
Eastern Europe | First Class | 658 | 2.00 | 1.00 | 1.00 | 96.05
Northern Europe | First Class | 1418 | 2.00 | 1.00 | 1.00 | 95.98
Central America | First Class | 4382 | 2.00 | 1.00 | 1.00 | 95.50
Southeast Asia | First Class | 1552 | 2.00 | 1.00 | 1.00 | 95.49
West Asia | First Class | 994 | 2.00 | 1.00 | 1.00 | 95.27
Western Europe | First Class | 4312 | 2.00 | 1.00 | 1.00 | 95.25
Result Grid
Form Editor
Field Types
Query Stats
Execution Plan
Result 8
Result 9
Result 10
Read Only
Output
Action Output
| Time | Action | Message
55 14:06:12 USE dataco_db 0 row(s) affected
56 14:06:12 SELECT order_region, shipping_mode, COUNT(order_id) AS total_orders, ROUND(AVG(days_for_shipping_real), 2) AS ... 15 row(s) returned
57 14:06:13 USE dataco_db 0 row(s) affected
58 14:06:13 SELECT category_name, COUNT(order_id) AS total_orders, SUM(CASE WHEN benefit_per_order < 0 THEN 1 ELSE 0 ... 10 row(s) returned
59 14:06:13 USE dataco_db 0 row(s) affected
60 14:06:13 SELECT market, order_status, COUNT(order_id) AS order_count, ROUND (SUM(sales), 2) AS impact_sales_usd From d... 10 row(s) returned


SQL File 8*
SQL File 4*
SQL File 3*
SQL File 5*
SQL File 6*
Limit to 50000 rows
34 market,
35 order_status,
36 COUNT(order_id) AS order_count,
37 ROUND (SUM(sales), 2) AS impact_sales_usd
38 From dataco_supply_chain
39 WHERE order_status IN('SUSPECTED FRAUD', 'CANCELED', 'PENDING_PAYMENT')
40 GROUP BY market, order_status
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Fetch rows:
category_name total_orders loss_making_orders loss_order_pct total_direct_losses_usd total_sales_usd
Fishing 17325 3209 18.52 -728570.95 6929653.50
Cleats 24551 4590 18.70 -452594.21 4431942.66
Camping & Hiking 13729 2590 18.87 -443082.23 4118425.42
Cardio Equipment 12487 2332 18.68 -402647.26 3694843.20
Water Sports 15540 2924 18.82 -334569.63 3113844.60
Women's Apparel 21035 3923 18.65 -323772.87 3147800.00
Men's Footwear 22246 4169 18.74 -309269.70 2891757.54
Indoor/Outdoor Games 19298 3617 18.74 -298637.48 2888993.94
Shop By Sport 10984 2154 19.61 -144310.56 1309522.02
Computers 442 73 16.52 -74967.13 663000.00
Result Grid
Form Editor
Field Types
Query Stats
Execution Plan
Result 8
Result 9
Result 10
Read Only
Output
Action Output
Time Action Message
55 14:06:12 USE dataco_db 0 row(s) affected
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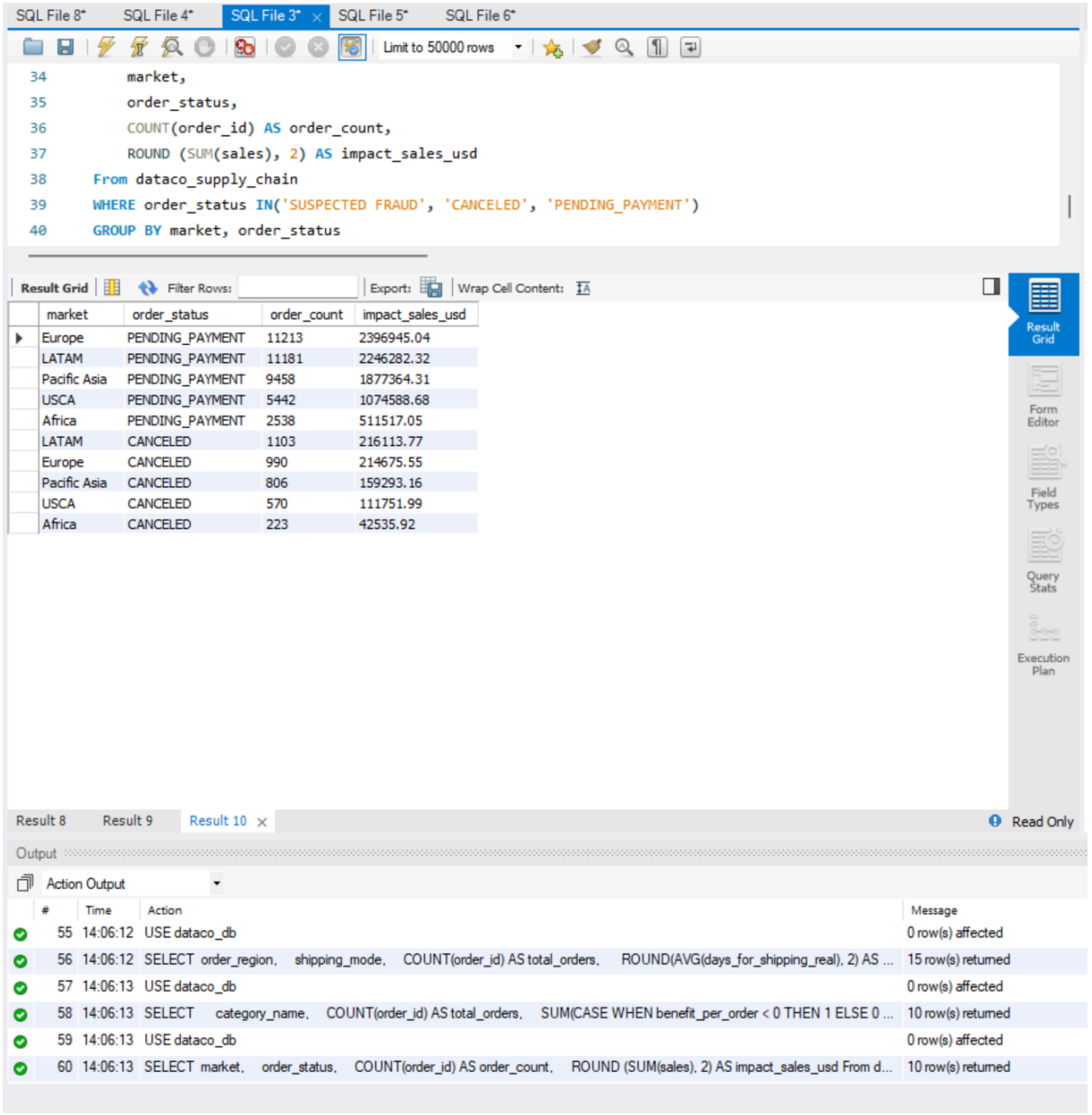
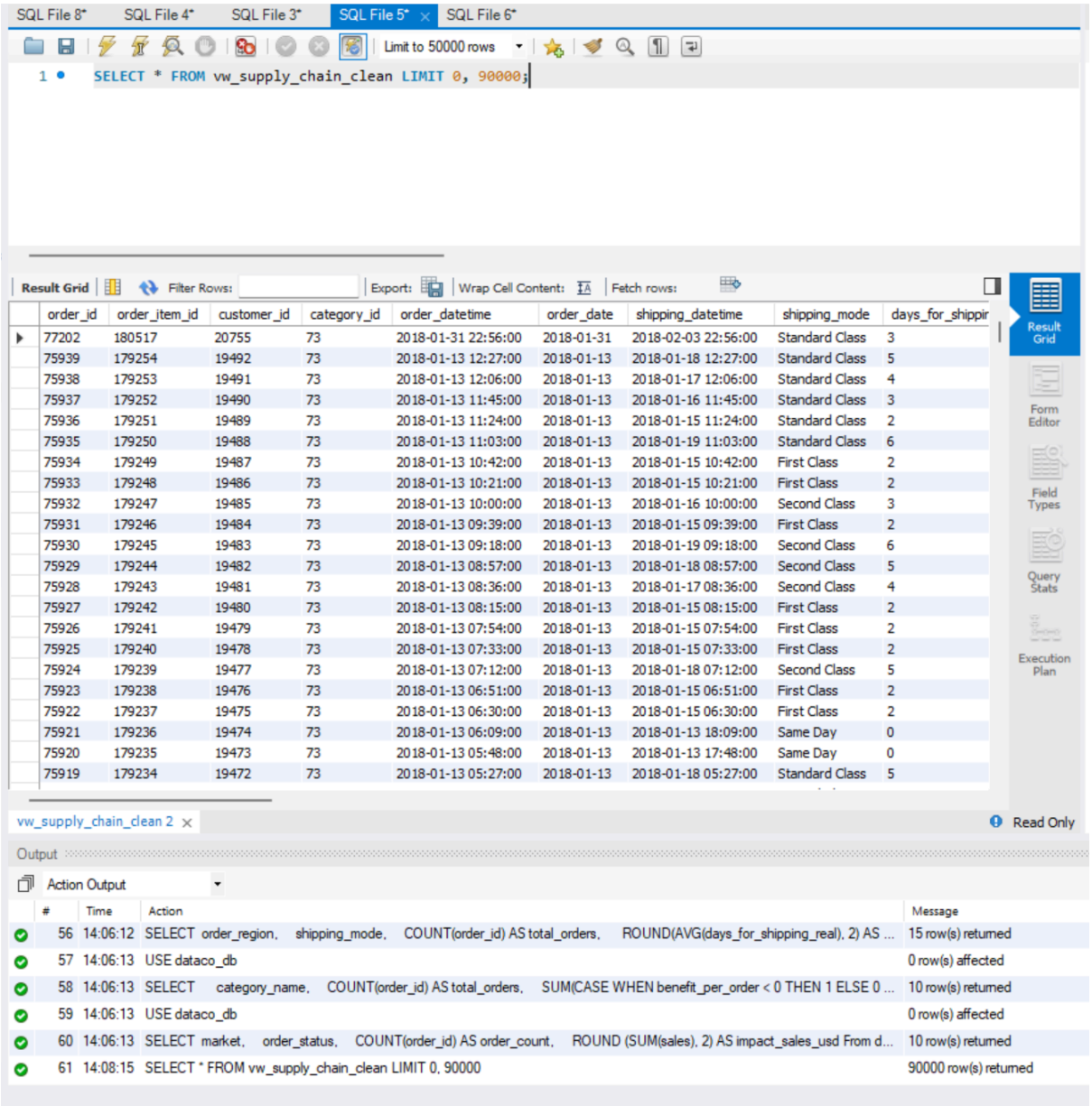
59 14:06:13 USE dataco_db 0 row(s) affected
60 14:06:13 SELECT market, order_status, COUNT(order_id) AS order_count, ROUND (SUM(sales), 2) AS impact_sales_usd From d... 10 row(s) returned



Large Dataset Batching Queries

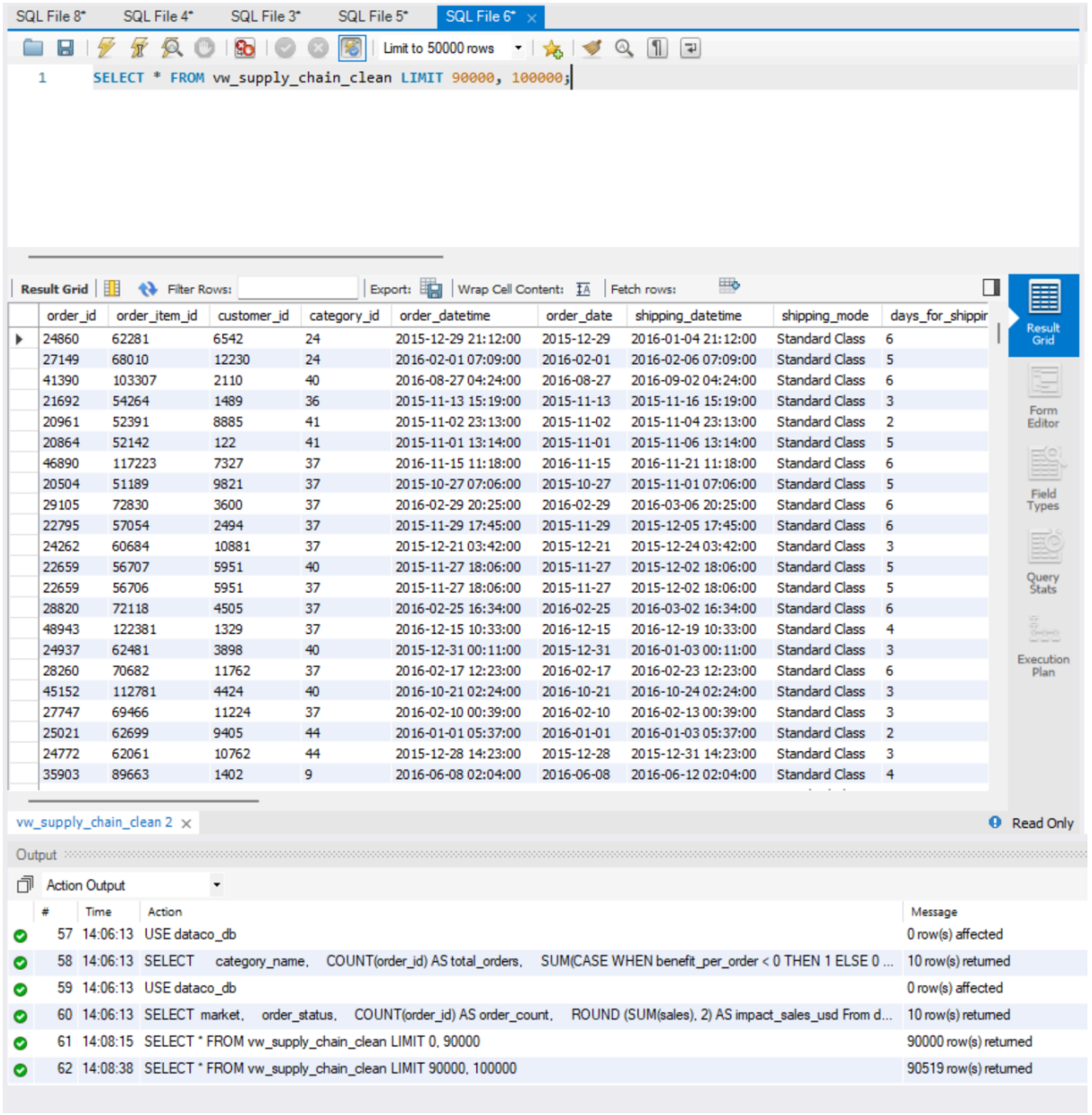
Batch Export: Records 1 to 90,000

```sql
SELECT * FROM vw_supply_chain_clean LIMIT 0, 90000;
```



Batch Export: Records 90,001 to 100,000

```sql
SELECT * FROM vw_supply_chain_clean LIMIT 90000, 100000;
```


SQL File 8*
SQL File 4*
SQL File 3*
SQL File 5*
SQL File 6*
Limit to 50000 rows
1 SELECT * FROM vw_supply_chain_clean LIMIT 90000, 100000;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Fetch rows:
order_id order_item_id customer_id category_id order_datetime order_date shipping_datetime shipping_mode days_for_shippir
24860 62281 6542 24 2015-12-29 21:12:00 2015-12-29 2016-01-04 21:12:00 Standard Class 6
27149 68010 12230 24 2016-02-01 07:09:00 2016-02-01 2016-02-06 07:09:00 Standard Class 5
41390 103307 2110 40 2016-08-27 04:24:00 2016-08-27 2016-09-02 04:24:00 Standard Class 6
21692 54264 1489 36 2015-11-13 15:19:00 2015-11-13 2015-11-16 15:19:00 Standard Class 3
20961 52391 8885 41 2015-11-02 23:13:00 2015-11-02 2015-11-04 23:13:00 Standard Class 2
20864 52142 122 41 2015-11-01 13:14:00 2015-11-01 2015-11-06 13:14:00 Standard Class 5
46890 117223 7327 37 2016-11-15 11:18:00 2016-11-15 2016-11-21 11:18:00 Standard Class 6
20504 51189 9821 37 2015-10-27 07:06:00 2015-10-27 2015-11-01 07:06:00 Standard Class 5
29105 72830 3600 37 2016-02-29 20:25:00 2016-02-29 2016-03-06 20:25:00 Standard Class 6
22795 57054 2494 37 2015-11-29 17:45:00 2015-11-29 2015-12-05 17:45:00 Standard Class 6
24262 60684 10881 37 2015-12-21 03:42:00 2015-12-21 2015-12-24 03:42:00 Standard Class 3
22659 56707 5951 40 2015-11-27 18:06:00 2015-11-27 2015-12-02 18:06:00 Standard Class 5
22659 56706 5951 37 2015-11-27 18:06:00 2015-11-27 2015-12-02 18:06:00 Standard Class 5
28820 72118 4505 37 2016-02-25 16:34:00 2016-02-25 2016-03-02 16:34:00 Standard Class 6
48943 122381 1329 37 2016-12-15 10:33:00 2016-12-15 2016-12-19 10:33:00 Standard Class 4
24937 62481 3898 40 2015-12-31 00:11:00 2015-12-31 2016-01-03 00:11:00 Standard Class 3
28260 70682 11762 37 2016-02-17 12:23:00 2016-02-17 2016-02-23 12:23:00 Standard Class 6
45152 112781 4424 40 2016-10-21 02:24:00 2016-10-21 2016-10-24 02:24:00 Standard Class 3
27747 69466 11224 37 2016-02-10 00:39:00 2016-02-10 2016-02-13 00:39:00 Standard Class 3
25021 62699 9405 44 2016-01-01 05:37:00 2016-01-01 2016-01-03 05:37:00 Standard Class 2
24772 62061 10762 44 2015-12-28 14:23:00 2015-12-28 2015-12-31 14:23:00 Standard Class 3
35903 89663 1402 9 2016-06-08 02:04:00 2016-06-08 2016-06-12 02:04:00 Standard Class 4
Result Grid
Form Editor
Field Types
Query Stats
Execution Plan
vw_supply_chain_clean 2
Read Only
Output
Action Output
Time Action Message
57 14:06:13 USE dataco_db 0 row(s) affected
58 14:06:13 SELECT category_name, COUNT(order_id) AS total_orders, SUM(CASE WHEN benefit_per_order < 0 THEN 1 ELSE 0 ... 10 row(s) returned
59 14:06:13 USE dataco_db 0 row(s) affected
60 14:06:13 SELECT market, order_status, COUNT(order_id) AS order_count, ROUND (SUM(sales), 2) AS impact_sales_usd From d... 10 row(s) returned
61 14:08:15 SELECT * FROM vw_supply_chain_clean LIMIT 0, 90000 90000 row(s) returned
62 14:08:38 SELECT * FROM vw_supply_chain_clean LIMIT 90000, 100000 90519 row(s) returned